Profit Loss Assignment

21 July 2025 21:39

1) CP MP SP

$100 \rightarrow 120 \xrightarrow{-\frac{1}{10}} 12\cancel{0}\times\frac{9}{\cancel{10}} = 84$

$120 \xrightarrow{\frac{1}{5}} \overset{24}{\cancel{120}}\times\frac{4}{\cancel{5}} = 96$

$12 = 4.8$

$1 = \frac{\cancel{4.8}^{4}}{\cancel{12}\times 10}$

$= 0.4$

$100 \times 0.4 = \underline{40}$ (Am) ⓓ

2)

OP CP SP

$\boxed{100} \rightarrow 90 \longrightarrow 9\cancel{0}\times\frac{12}{1\cancel{0}} = \boxed{108}$

$\frac{8}{100}\times 100$

$= \underline{8\%}$ ⓓ (Am)

③

26×20
$= 520$

$x\times 30$
$30x$

$39(26+x)$

$$(520+30x)\times\frac{\overset{3}{\cancel{150}}}{\underset{2}{\cancel{100}}} = 39(26+x)$$

$\Rightarrow 1560 + 90x = 2(1014 + 39x)$
$= 2028 + 78x$

$39 \times 26 = 1014$

$90x - 78 = 12x$

$\Rightarrow 12x = 2028 - 1560 = 0468$

$\boxed{x = 39}$ ⓑ (Am)

④

$CP = x \quad SP = 1620$
$\downarrow$
$21x$

↳ Profit $= 1620 - 21x$

$1620 - 21x = 6x$

$\Rightarrow \overset{3}{\cancel{27}}x = \overset{\cancel{180}\ 60}{\cancel{1620}}$

$\boxed{x = 60}$ ⓓ

⑤ Ayush ⟶ Rahul

$12000 \longrightarrow 12\cancel{000}\times\frac{80}{1\cancel{00}}$

$= 9600 \longrightarrow 96\cancel{00}\times\frac{120}{1\cancel{00}}$

$= 11520$ ⓒ (Am)

⑥ discount $= 25 + 24 - \frac{\overset{6}{\cancel{24}}\times 25}{\underset{4}{\cancel{100}}}$

$= \underline{43\%}$ ⓑ

⑦

$x\times 25 \qquad 30\times 30$

$30(x+30)$

$$(25x + 900)\times\frac{11}{10} = 30(x+30)$$

$\Rightarrow 275x + 9900 = 300x + 9000$

$\Rightarrow 25x = 900$

$\boxed{x = 36\text{ kg}}$ ⓒ

25)900(36
75
150

⑧ $33\frac{1}{3}\% = \frac{1}{3} \qquad 16\frac{2}{3}\% = \frac{1}{6}$

CP MP SP

$\boxed{4500} \quad \overset{1500}{\cancel{4500}}\times\frac{4}{\cancel{3}}$

$= 6000 \times \frac{5}{6} = \boxed{5000}$

Profit $= \underline{500}$ (Am) ⓒ

⑨ Discount $= 20 + 25 - 5$

$= 40\%$

$MP \times \frac{6}{10} = \overset{3000}{\cancel{18000}}$

$\therefore MP = \underline{30000}$ ⓐ (Am)

⑩

CP. SP

$100 \longrightarrow 110$

$95 \longrightarrow \frac{\overset{19}{\cancel{95}}\times\overset{6}{\cancel{120}}}{\underset{2}{\cancel{100}}} = 114$

$4 = 80$
$1 = 20$

$\therefore$ Cost Price $= 20\times 100$
$= \underline{2000}$ (Am) ⓒ

Set - 02

① CP

$15 \rightarrow ₹1$
$1 \rightarrow \frac{1}{15}$

SP
↳ $\frac{1}{12}$

$\therefore \frac{CP}{SP} = \frac{\frac{1}{15}}{\frac{1}{12}} = \frac{12}{15} = \frac{4}{5}$

$\therefore$ gain $= \frac{1}{\cancel{4}}\times\cancel{100} = \underline{25\%}$ (Am)

②

Discount $= \frac{\text{Free}}{\text{Total}}\times 100$ ✓

$= \frac{1}{\cancel{5}}\times\overset{20}{\cancel{100}}$

$= 20\%$

$\therefore$ No profit.

③ CP
↳ ₹1/g

$CP = 900$
$SP = 1000$

$\therefore$ Profit $= \frac{100}{900}\times 100$
$= 11.11\%$ (Am)

④ $CP_1 + CP_2 = 1008$

↳ $SP_1 = SP_2$

$\Rightarrow CP_1 \times \frac{\overset{4}{\cancel{8}}}{\underset{5}{\cancel{10}}} = CP_2 \times \frac{\overset{\overset{18}{\cancel{72}}}{\cancel{144}}}{\underset{\underset{10}{\cancel{50}}}{\cancel{100}}}$

$\Rightarrow \frac{CP_1}{CP_2} = \frac{18}{10} = \frac{9}{5}$

$\therefore CP_1 = \overset{72}{\cancel{1008}}\times\frac{9}{\cancel{14}}$

$= \underline{648}$ (Am)

14)1008(72
98
28
28

⑤ $CP_1 + CP_2 = 5000$

↳ No Profit, No loss

$\therefore$ Total CP = Total SP = 3000

$\Rightarrow SP_1 + SP_2 = 3000$

$\frac{122CP_1}{100} + \frac{92CP_2}{100} = CP_1 + CP_2$

$\Rightarrow \frac{22CP_1}{\cancel{100}} = \frac{8CP_2}{\cancel{100}}$

$\frac{CP_1}{CP_2} = \frac{8}{22} = \frac{4}{\boxed{11}}$

$CP_2 = \overset{200}{\cancel{3000}}\times\frac{11}{\cancel{15}}$

$= \underline{2200}$ (Am)

⑥ CP MP SP

$\boxed{300}$

$3\cancel{00}\times\frac{88}{\cancel{100}} = \boxed{264}$

$CP\times\frac{11\cancel{0}}{10\cancel{0}} = \overset{24}{\cancel{264}}$

$\Rightarrow CP = \boxed{240}$

$\therefore$ Profit $= \frac{\cancel{60}}{\underset{4}{\cancel{240}}}\times\overset{25}{\cancel{100}}$

$= \underline{25\%}$ (Am)

⑦

SP: Shirt + T-shirt $= \boxed{2940}$

$CP\times\frac{\overset{21}{\cancel{105}}}{\underset{20}{\cancel{100}}} = \overset{140}{\cancel{2940}}$

$\therefore CP = \underline{2800}\ (CP_S + CP_T)$

21)2940(14
21
84

$SP_S + SP_T = 2940$

$CP_S\times\frac{120}{100} + CP_T\times\frac{85}{100} = 2940$

↳ $(CP_S + CP_T)\times\frac{105}{100}$.

$\Rightarrow 120CP_S + 85CP_T = 105CP_S + 105CP_T$

$\Rightarrow \overset{3}{\cancel{15}}CP_S = \overset{4}{\cancel{20}}CP_2$

$\frac{CP_S}{CP_T} = \frac{4}{3}$

$\therefore CP_S = \overset{400}{\cancel{2800}}\times\frac{4}{\cancel{7}}$

$= \underline{1600}$ (Am)

8)

CP $\quad SP = 700$

$\therefore$ Profit $= 700 - CP$ | Profit $= 650 - CP$

$14\frac{2}{7}\% = \frac{1}{7}$

$(700 - CP) = (650 - CP)\times\frac{8}{7}$

$\Rightarrow 4900 - 7CP = 5200 - 8CP$

$\Rightarrow CP = \underline{300}$

$\therefore SP = 3\cancel{00}\times\frac{125}{\cancel{100}}$

$\boxed{SP = 375}$ (Am) ⓓ